Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 25

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1	Index rules with positive indices, Index rules with negative indices	2	U235,U694
3	Divide integers, Find percentages of amounts, Percentage change	5	U453,U554, U773
5a	Calculating with speed, Estimating calculations	3	U151,U225
5b	Estimating calculations	1	U225
7	Find the area of simple shapes, Find the area of circles, Construct fractions	4	U993,U950
9a	Interpreting cumulative frequency graphs, Drawing box plots	3	U642,U879
9b	Comparing populations using box plots and cumulative frequency graphs	2	U507
9c	Comparing populations using box plots and cumulative frequency graphs	1	U507
11	Solving direct proportion word problems, Percentage change without a calculator	3	U721,U773
13	Combining transformations	2	U766
15	Finding the surface area and volume of similar shapes	4	U110
17	Finding the turning point of a quadratic graph by completing the square	4	U769
19a	Substituting into composite functions	2	U895
19b	Finding inverse functions	2	U996
21	Solving geometric problems using vectors	5	U781
	Total	43	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Work out the value of $\dfrac{3^7 \times 3^{-2}}{3^3}$

.....

(Total for Question 1 is 2 marks)

3 A bonus of £2100 is shared by 10 people who work for a company.
40% of the bonus is shared equally between 3 managers.
The rest of the bonus is shared equally between 7 salesmen.

One of the salesmen says,

"If the bonus is shared equally between all 10 people I will get 25% more money."

Is the salesman correct?
You must show how you get your answer.

(Total for Question 3 is 5 marks)

5 A plane travels at a speed of 213 miles per hour.

(*a*) Work out an estimate for the number of seconds the plane takes to travel 1 mile.

..... seconds

(3)

(*b*) Is your answer to part (*a*) an underestimate or an overestimate?
Give a reason for your answer.

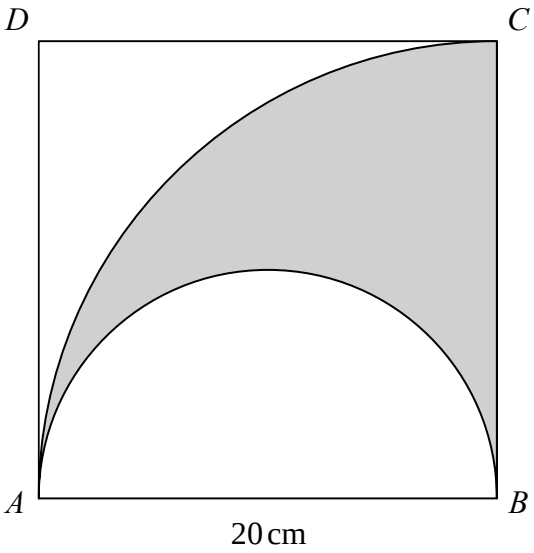
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(1)

(Total for Question 5 is 4 marks)

7 The diagram shows a square *ABCD* with sides of length 20 cm.
It also shows a semicircle and an arc of a circle.



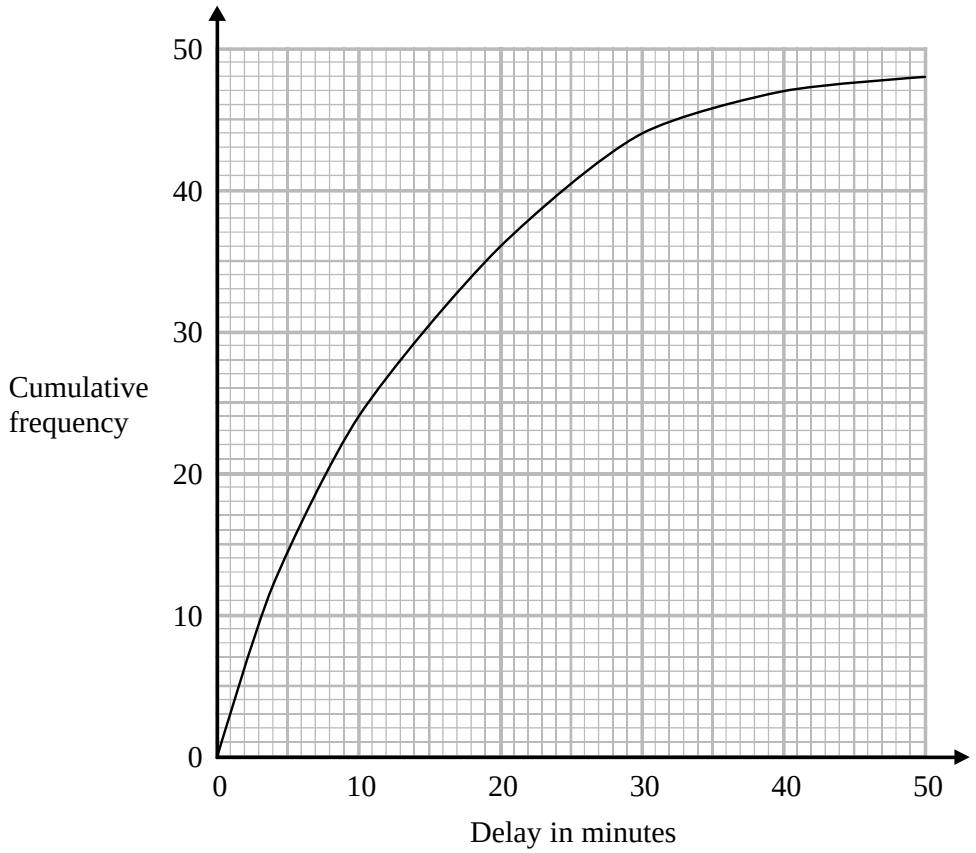
AB is the diameter of the semicircle.
AC is an arc of a circle with centre *B*.

Show that $\dfrac{\text{area of shaded region}}{\text{area of square}} = \dfrac{p}{8}$

(Total for Question 7 is 4 marks)

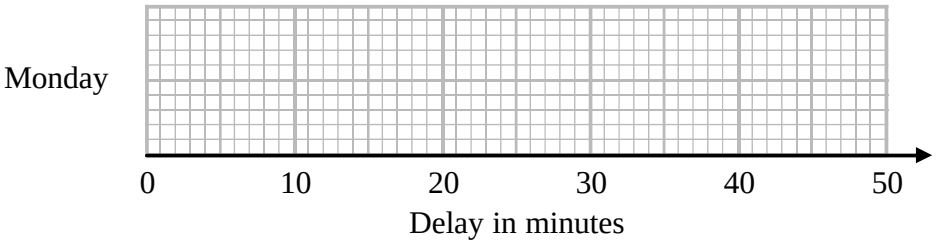
9 The times that 48 trains left a station on Monday were recorded.

The cumulative frequency graph gives information about the numbers of minutes the trains were delayed, correct to the nearest minute.



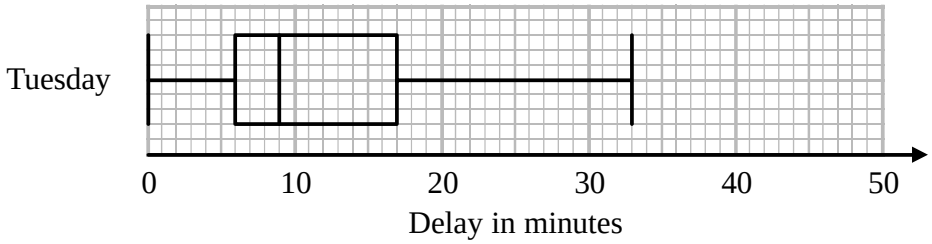
The shortest delay was 0 minutes.
The longest delay was 42 minutes.

(*a*) On the grid below, draw a box plot for the information about the delays on Monday.



(3)

48 trains left the station on Tuesday.
The box plot below gives information about the delays on Tuesday.



(*b*) Compare the distribution of the delays on Monday with the distribution of the delays on Tuesday.

.....

.....

.....

.....

(2)

Mary says,

"The longest delay on Tuesday was 33 minutes.
This means that there must be some delays of between 25 minutes and 30 minutes."

(*c*) Is Mary right?
You must give a reason for your answer.

.....

.....

(1)

(Total for Question 9 is 6 marks)

11 Jack and Sadia work for a company that sells boxes of breakfast cereal.

The company wants to have a special offer.

Here is Jack's idea for the special offer.

Put 25% more cereal into each box and do **not** change the price.

Here is Sadia's idea.

Reduce the price and do **not** change the amount of cereal in each box.

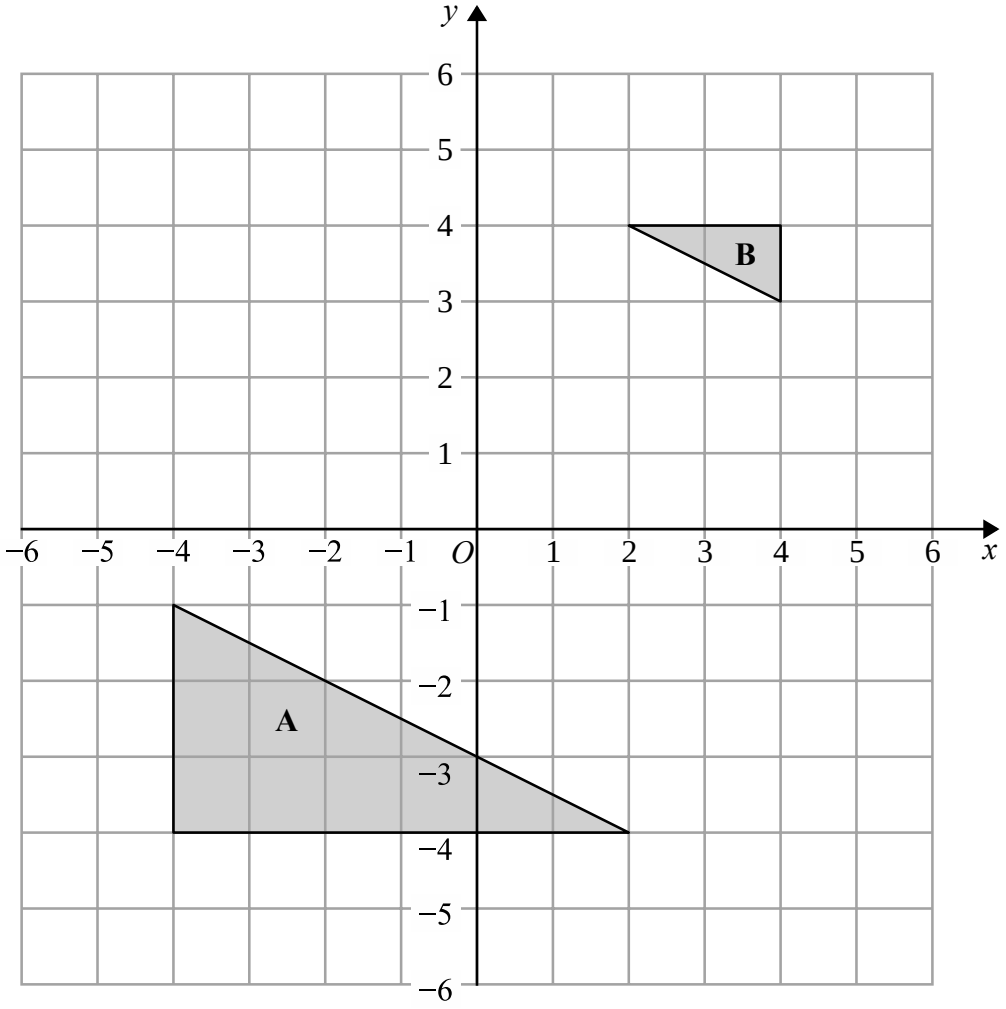
Sadia wants her idea to give the same value for money as Jack's idea.

By what percentage does she need to reduce the price?

..... %

(Total for Question 11 is 3 marks)

13



Describe fully the single transformation that maps triangle **A** onto triangle **B**.

.....

(Total for Question 13 is 2 marks)

15 Three solid shapes **A**, **B** and **C** are similar.

The surface area of shape **A** is 4 cm^2
The surface area of shape **B** is 25 cm^2

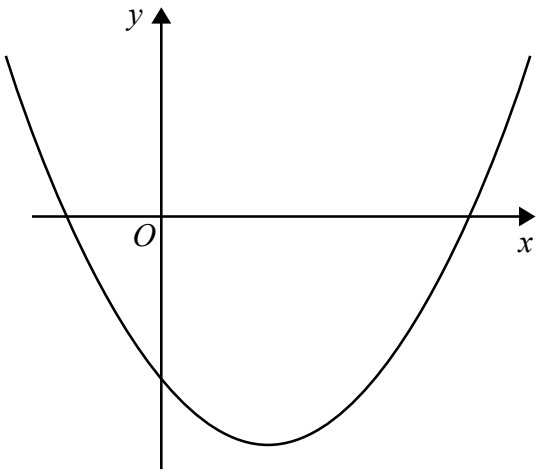
The ratio of the volume of shape **B** to the volume of shape **C** is 27 : 64

Work out the ratio of the height of shape **A** to the height of shape **C**.
Give your answer in its simplest form.

.....

(Total for Question 15 is 4 marks)

17 Here is a sketch of a curve.



The equation of the curve is $y = x^2 + ax + b$ where a and b are integers.

The points $(0, -5)$ and $(5, 0)$ lie on the curve.

Find the coordinates of the turning point of the curve.

(..... , ...)

(Total for Question 17 is 4 marks)

19 For all values of x

$$f(x) = (x + 1)^2 \quad \text{and} \quad g(x) = 2(x - 1)$$

(*a*) Show that $gf(x) = 2x(x + 2)$

(2)

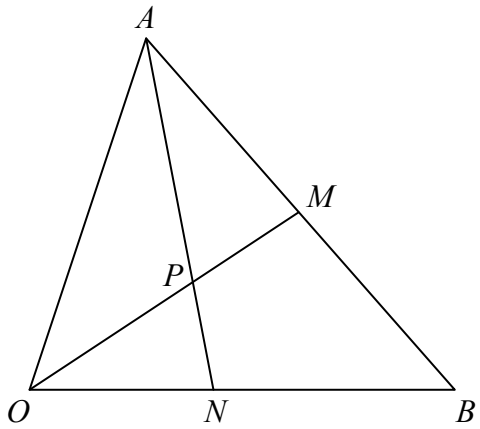
(*b*) Find $g^{-1}(7)$

.....

(2)

(Total for Question 19 is 4 marks)

21



OAB is a triangle.
OPM and *APN* are straight lines.
M is the midpoint of *AB*.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

OP : *PM* = 3 : 2

Work out the ratio *ON* : *NB*

.....

(Total for Question 21 is 5 marks)

TOTAL FOR PAPER IS 43 MARKS